

# Tong Ye (Niklaus) Zhang

GAMES RESEARCH · GAME DESIGN · TECHNICAL ART

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## PROFILE

Games researcher and designer bringing research, design and hands-on tool experience to technical art. My research examines how developers and players negotiate difficulty and time, through a master's thesis at the University of Toronto and a peer-reviewed paper with the Canadian Game Studies Association, and has since been developed into the Difficulty Design Vault, a teaching tool for game-design courses. As a designer, I have created a narrative action demo and a competitive and co-operative board game, and I mentor students from initial concept to finished projects in Unreal Engine, Maya and 3ds Max. More than 40,000 hours of play underpin a lasting interest in how games are built, most recently applied to reproducing the rendering style of *Return of the Obra Dinn* as a real-time shader in Unity. Native speaker of English, Cantonese and Mandarin.

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## SELECTED WORK

**Student Portfolio Mentoring** · Unreal Engine, Houdini, TouchDesigner, 3ds Max, Raspberry Pi, Arduino      Ongoing

- Guide study-abroad applicants through every stage of a game or animation portfolio, from topic selection to final layout. Students carry out all production work; each tool is taught as the project requires it.
- *Hallucination*: first-person narrative puzzle game in Unreal Engine, with a clue board, inspectable items and code-lock puzzles implemented in Blueprints.
- *Sigil of Catharsis*: interactive virtual scene with characters modelled in Maya and a strike interaction driving Niagara particles, material changes and lighting transitions in Unreal Engine.
- *Greed*: short animation with landscape and underwater materials built in Unreal Engine, and props modelled in 3ds Max and textured in Substance Painter.

**Reproducing the *Return of the Obra Dinn* Dither** · Unity URP, HLSL, JavaScript      2026

- Reverse-engineered the game's 1-bit rendering style through visual analysis, arriving at an 8x8 ordered Bayer matrix, then validated it against the developer's published devlog and identified the missing step: an approximately 10% contrast increase before thresholding.
- Implemented the result as a live side-by-side browser demonstration and as a real-time post-process shader in Unity, ported to HLSL and integrated through the URP render graph.

**Difficulty Design Vault** · independent project      2026

- Designed and built a bilingual (English and Chinese) teaching tool for game-design courses. Student analyses of difficulty mechanics are coded against a public research framework and published as evidence-graded design guidance for small development teams, supported by a mechanism library, a recommendation tool and a public claims registry.

**Decypress** · narrative action demo, Unity      2021

- Wrote the story, lore and dialogue, composed the original soundtrack, and designed collectibles, progression triggers and environmental storytelling across four levels, in collaboration with Scott Warren (SayinCraft Studios), who led programming and visuals.

**Dive Down the Dungeon Deep** · competitive and co-operative board game      2023

- Led a five-person team as project manager and core mechanic designer, defining the rules, board layout, and shared and individual win conditions; revised the action economy and turn pacing after two rounds of playtesting with 36 players.

## RESEARCH

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**Here Be Lions** · Master's thesis, University of Toronto 2025  
*Reframing Challenge Design and Difficulty Adjustment as Negotiation in Video Games*

- Synthesized 8 developer interviews, discourse analysis of 590+ player comments and comparative case studies of 35 games into the Negotiated Difficulty Framework, which now underpins the Difficulty Design Vault.

**Captive Audience** · Peer-reviewed paper, Canadian Game Studies Association 2023  
*A Study in Time and Player Experience in Online and Paper Magic: The Gathering*

- Field study of 42 players across 200+ games at 5 locations, identifying competitiveness as a key modulator of flow.

## EXPERIENCE

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**UX/UI Engineer** · China Telecom Global, Hong Kong Jul 2026 – Present

- Research and design the interfaces for five concurrent applications for the Hong Kong Fire Services Department and Transport Department, including administrative systems, an intranet, member-facing apps and internal tools.
- Coordinate front-end delivery across these applications in a de facto product-management capacity, from user research and interface design through developer handoff and client review.
- Produce build-ready HTML prototypes for React and Flutter developers and remain involved through implementation, repository review and quality assurance.
- Authored close to 1,000 audit-ready UAT test cases to government SLC standards, and provided onsite delivery at a client data centre under restricted tooling.

**Teaching Assistant** · University of Toronto 2023 – 2024

- Mentored 12 student teams through playtesting and Figma-based UX research, provided written feedback and thematic analysis on 120+ reports, and led weekly game-design tutorials for students from mixed backgrounds.

**Localization & Creative Direction Associate** · Oak Smart Advertising Media (part-time) 2018 – 2025

- As assistant creative director, led trilingual campaigns for clients including China Unicom Hong Kong, coordinating online content with offline events from planning through delivery.

## EDUCATION

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**Master of Information** · University of Toronto 2025  
*Culture & Technology and User Experience Design concentrations, thesis pathway*

**Bachelor of Arts** · University of Toronto Mississauga, joint program with Sheridan College 2023  
*Communication, Culture, Information & Technology; Certificate in Digital Communications, Sheridan College*

## SKILLS

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| <b>Engines &amp; 3D</b>      | Unity, Unreal Engine, Godot, Houdini, 3ds Max, Substance Painter, TouchDesigner                                                                                                       |
| <b>Physical computing</b>    | Raspberry Pi, Arduino                                                                                                                                                                 |
| <b>Programming &amp; web</b> | Python, JavaScript, SQL, HTML, CSS, React, Flutter                                                                                                                                    |
| <b>Design &amp; research</b> | UX research, playtesting, Figma, FigJam, Photoshop, Illustrator                                                                                                                       |
| <b>Project delivery</b>      | Agile / Scrum, Jira, Git, GitHub                                                                                                                                                      |
| <b>Languages</b>             | English, Cantonese and Mandarin (native); Japanese (conversational)                                                                                                                   |
| <b>Gaming</b>                | 40,000+ hours across genres. Competitive: Guilty Gear Strive / Street Fighter 6 peak global rank #142; Hearthstone Legend; MTG Grand Prix Vancouver Day 2 (top 15%); MTG Arena Mythic |